

DevOps

Curriculum



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Phase 1: Linux Foundations (Core for DevOps)

Module 1: Linux Fundamentals & Getting Comfortable

Module 2: Essential Commands & Text Processing

Module 3: Users, Permissions & Security Basics

Module 4: Process & Resource Management

Module 5: Package Management & Software Installation

Module 6: Networking Essentials for DevOps

Module 7: Storage & Disk Management

Module 8: Shell Scripting & Automation Basics

Module 9: Systemd & Service Management

Phase 2: Source Control Management

Module 10: Git & GitHub

- Introduction to Source Control & DevOps relevance
- Git Commands: init, clone, add, commit, status, branch, checkout, merge, pull, push, log, stash, revert
- GitHub: Repositories, Forking, Pull Requests, Merging
- Branch Protection, Issues
- GitHub Actions (Basic CI/CD)

Phase 3: Build & Dependency Management

Module 11: Maven (Java Build Tool)

- Maven Overview & Importance
- POM, Dependencies, Plugins
- Maven Commands: clean, compile, package, install
- Hands-on:
 - Java project setup
 - Dependency management
 - Unit testing with Maven

Phase 4: Continuous Integration & Delivery

Module 12: Jenkins

- CI/CD Concepts
- Jenkins Installation & Setup
- Jobs: Freestyle & Pipeline
- Triggers: Poll SCM, Webhooks
- Post-build Actions: Email, Artifact Archiving
- Jenkins Pipelines: Declarative & Scripted
- Plugin Management
- Hands-on:
 - Build automation
 - GitHub integration

Phase 5: Containerization

Module 13: Docker

- Introduction to Containers & Docker
- Docker Commands: pull, build, run, ps, stop, rm
- Writing Dockerfiles
- Managing Images, Volumes & Networks
- Hands-on:
 - Build & run containers

Phase 6: Configuration Management

Module 14: Ansible

- Introduction to Ansible in DevOps
- Core Concepts: Inventory, Playbooks, Roles, Modules
- Commands: ansible, ansible-playbook, ansible-galaxy, ansible-vault
- Writing Playbooks
- Server Configuration & Automation
- Ansible with Jenkins for deployments

Phase 7: Container Orchestration

Module 15: Kubernetes

- Kubernetes Overview & Architecture
- Local Cluster Setup (Minikube / Docker Desktop)
- Core Components: Pods, Services, Deployments
- kubectl Commands: create, apply, get, describe, delete

- Hands-on:
 - Application deployment
 - Service exposure
 - Scaling & rollout strategies

Phase 8: Monitoring & Observability

Module 16: Prometheus Fundamentals & Architecture

Module 17: Installation & Basic Configuration

Module 18: Prometheus Data Model & Metrics Types

Module 19: PromQL Basics – Querying Metrics

Module 20: Advanced PromQL – Functions, Operators & Aggregations

Module 21: Exporters & Instrumentation

Module 22: Service Discovery & Dynamic Targets

Module 23: Recording Rules & Alerting Rules

Module 24: Alertmanager – Routing, Grouping & Notifications

Module 25: Grafana Integration – Dashboards & Visualization

Module 26: Monitoring Kubernetes & Containers

Module 27: Production Best Practices, HA & Troubleshooting

Phase 9: End-to-End DevOps Projects

Workflow 1:

- Git → Maven → Jenkins → Tomcat

Workflow 2:

- Git → Maven → Jenkins → Ansible → Docker

Workflow 3:

- Git → Maven → Jenkins → Ansible → Kubernetes

Outcome

- Strong Linux & automation foundation
- Real-time CI/CD & container orchestration
- Production-grade monitoring with Prometheus & Grafana
- Job-ready DevOps skillset with hands-on projects

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